Lecture 2: Data and Describing Data

Lecturer

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CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

AGENDA

You have 4,000 rows of data. Your director gives you 30 seconds. What do you say?

WHY COMPRESS AT ALL

Why no single cell in the file answers the director's question.

WHAT THIS DATA IS

Observation unit, three structures, measurement scales.

THE CENTRE

Mean, median, mode — and why economic variables lean right.

SPREAD AND POSITION

SD, CV, IQR, percentiles, z-scores.

HANDS-ON

The professional table, the tools, and the file Session 3 needs.

WHY COMPRESS AT ALL

Same file, two questions

Q1 — What was Vinamilk's revenue in 2023?

A **lookup**: one row, one column, one cell.

Q2 — How did Vietnam's listed firms perform in 2023?

A **summary**: no single cell answers it.

Raw data answers lookups. Statistics exist to answer summaries.

WHY COMPRESS AT ALL

A statistic is a compression

The shape every topic in this session follows

A rule that keeps some, discards the rest

The mean of 20,000 revenues uses the **magnitude** of every single value — and discards which firm had which value, and all ordering. This is a huge loss of information

So why we have to compress information?

Because 20,000 lines of data is heavy to carry and interpret. No one can fully understand all numbers. We need something easier to interpret, but still keep a lot of information from the original object.

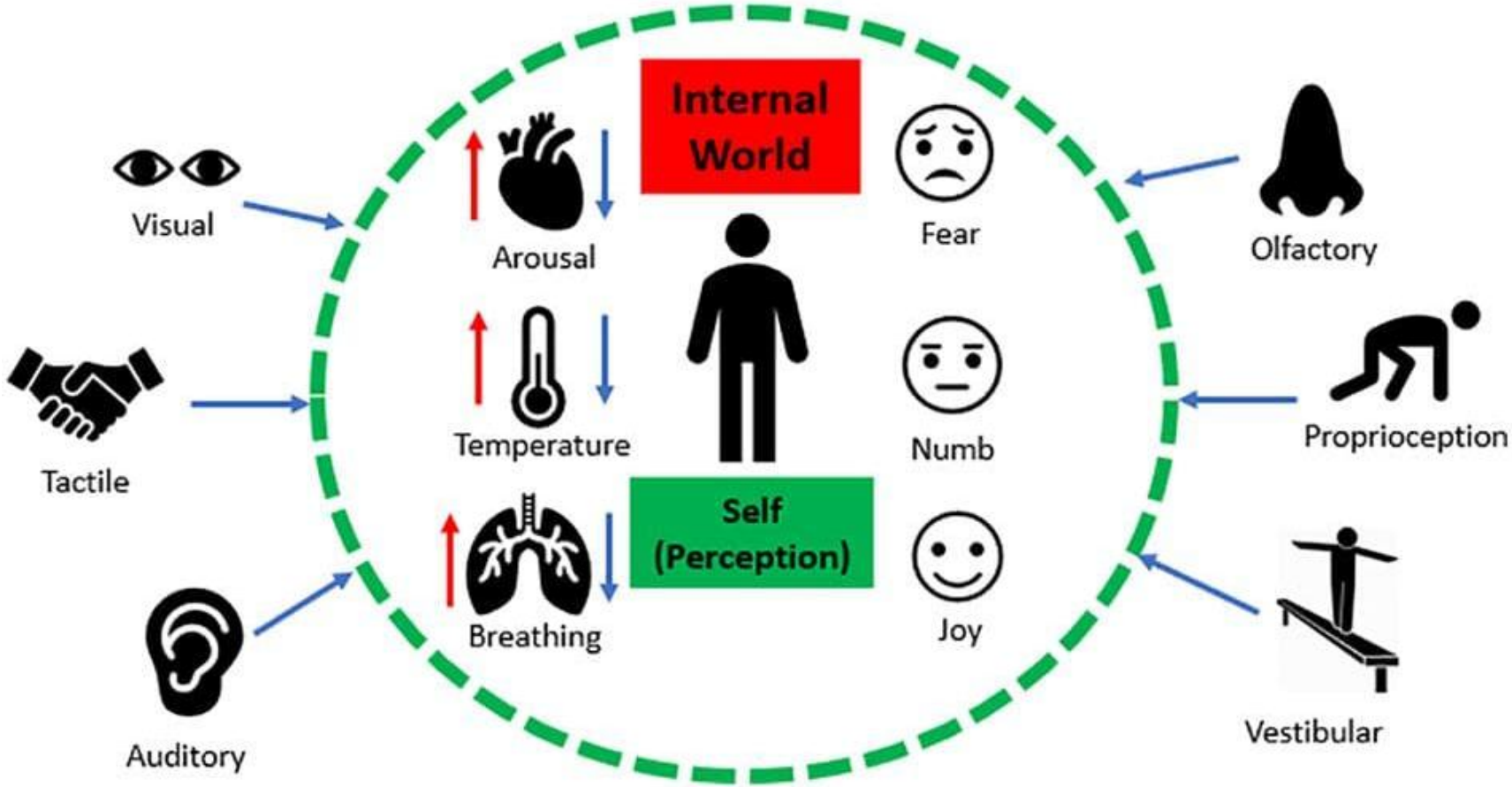
Misinformation is not a defect. It is the price of the compression.

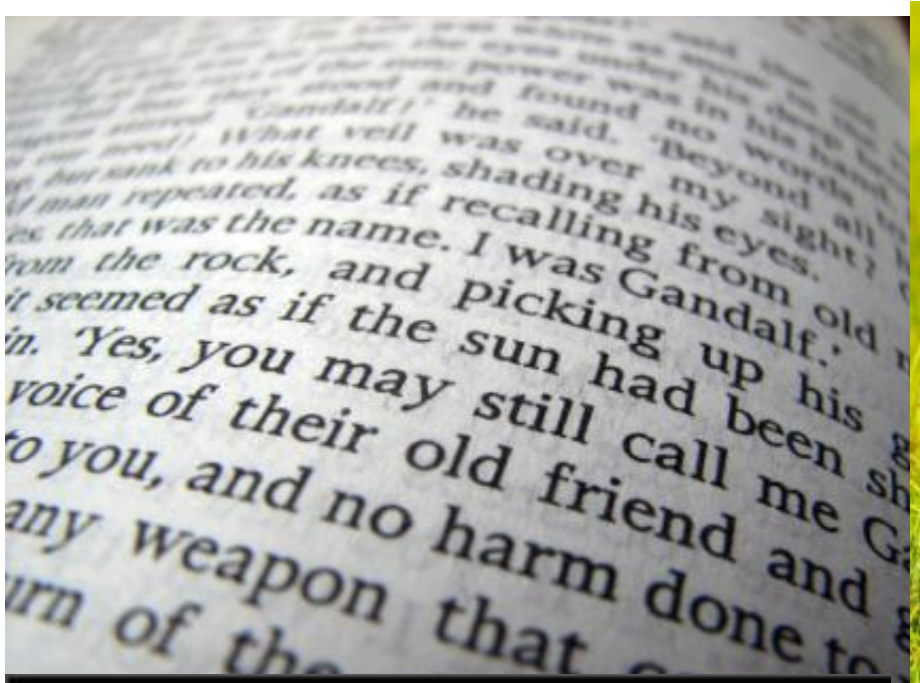
The skill is knowing which price you paid.

WHAT THIS DATA IS

Different Classification of Data

Think off how we perceive new information in daily activities







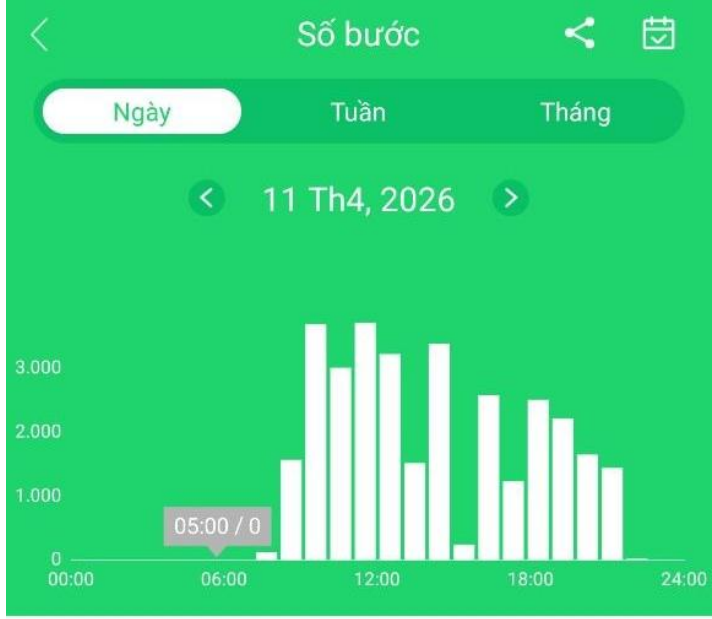


A	B	C	D	E	F	G	H	I	J
				UK		USA		Total	
Order Date	OrderID	Salesperson		Units	Order Amount	Units	Order Amount	Units	Order Amount
1/01/2021	10392	Fuller				13	1440	13	1440
2/01/2021	10397	Gloucester		17	716.72			17	716.72
2/01/2021	10771	Bromley		18	344			18	344
3/01/2021	10393	Finchley				16	2556.95	16	2556.95
3/						10	442	10	442
3/				9	2122.92			9	2122.92
6/01/2021	10396	Finchley						7	1903.8
8/01/2021	10399	Callahan				17	1765.6	17	1765.6
8/01/2021	10404	Fulle				7	1591.25	7	1591.25
9/01/2021	10398	Fulle				11	2505.6	11	2505.6
9/01/2021	10403	Coghill		18	855.01			18	855.01
10/01/2021	10401	Finchley				7	3868.6	7	3868.6
10/01/2021	10402	Bromley				11	2713.5	11	2713.5
13/01/2021	10406	Rayleigh		15	1830.78			15	1830.78
14/01/2021	10408	Callahan				10	1622.4	10	1622.4
14/01/2021	10409	Farnham		19	319.2			19	319.2
15/01/2021	10410	Farnham				16	802	16	802
15/01/2021	10412	Callahan				8	334.8	8	334.8

Column headings span two or more rows.

Total column(s).

Blank column for formatting.



Số bước	Khoảng cách	Kalo
32136	23.75 km	968.5 kcal

Dữ liệu hôm nay（18）　Nguồn khác*

Đi bộ 22:04~22:23	21	0.02 km	0.6 kcal
Chạy bộ 21:14~21:16	18	0.02 km	1.1 kcal
Đi bộ 21:14~21:33	1426	1.05 km	42.9 kcal
Đi bộ 20:00~20:33	1653	1.22 km	49.8 kcal

WHAT THIS DATA IS

Data Classification by Structure

Type	Structure	Unit	Machine-Representation
Tabular	Structured	Row, Record	2D Grid
Text	Unstructured	Token, words	Vector embedding
Image	Unstructured	Pixel	RGB Matrix
Video	Unstructured	Frame and Pixel	Sequence of frames
Waveform	Unstructured	Hz	Voice, Spectrogram

WHAT THIS DATA IS

Tabular Data

Key Characteristics

- **Rows**: Represent unique samples, entities, or chronological events
- **Columns**: Represent specific features, categories, or measured properties
- **Homogeneity**: Data types within a single column are typically uniform (e.g., all numbers, text, or dates)

Common Formats

- Spreadsheets: XLSX, ODS (used in Microsoft Excel or Google Sheets)
- Flat Files: CSV (Comma-Separated Values) and TSV (Tab-Separated Values)
- Databases: Relational database tables (like SQL)
- Web Formats: JSON, XML

The main modality of data for Statistics

WHAT THIS DATA IS

Panel Data

The most common type of Data in Economics and Econometrics

Slice	What you keep	Name
All firms, one year	many units, one time point	cross-section
One firm, all years	one unit, many time points	time series
All firms, all years	many units, many time points	panel

3 Dimensions: Who, When, What

WHAT THIS DATA IS

What happens when you pool the panel

One mean over two kinds of variation

Take the whole FirmSize column and compute one mean. It averages **between** firms and **within** firms at once. The formula sees 4,000 numbers, not 400 firms.

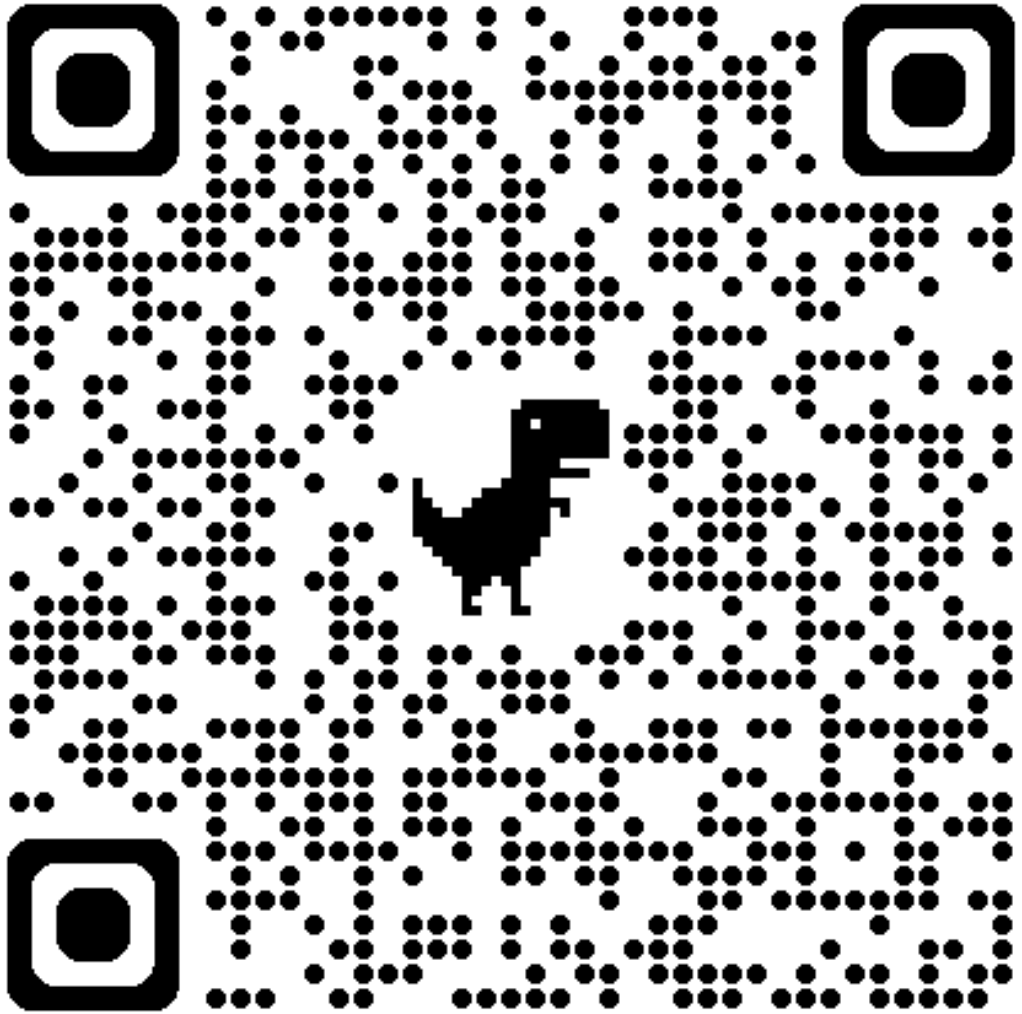
Rows are not independent

The same firm appears multiple times, so the rows are not independent

BankID	symbol	Year	exchang	organ_n	FirmAge	FirmSize	ROA
VNM	VNM	2025	HOSE	CTCP Sữa	49	5.33123E+13	0.1765107
VNM	VNM	2024	HOSE	CTCP Sữa	48	5.50491E+13	0.1706170
VNM	VNM	2023	HOSE	CTCP Sữa	47	5.26734E+13	0.1684686
VNM	VNM	2022	HOSE	CTCP Sữa	46	4.84827E+13	0.1756509
VNM	VNM	2021	HOSE	CTCP Sữa	45	5.33324E+13	0.1974873
VNM	VNM	2020	HOSE	CTCP Sữa	44	4.84325E+13	0.2291630
VNM	VNM	2019	HOSE	CTCP Sữa	43	4.46999E+13	0.236716
VNM	VNM	2018	HOSE	CTCP Sữa	42	3.73661E+13	0.2737047

WHAT THIS DATA IS

Take some time to complete this survey!



WHAT THIS DATA IS

Data Classification by Nature

This is classification from the perspective of Mathematics and Research: Numbers or Non-numbers

Variable	Quantitative	Qualitative	
No structure $=$ $\neq$		Color, Marital status,	Nominal
Ordinal Structure $=$ $\neq$ $\geq$	Age, Shoe size	Rank, Resemblance	Ordinal, Discrete
Ordinal ring Structure $=$ $\neq$ $\geq$ $+$ $\times$	Income, Length		Continuous

WHAT THIS DATA IS

Data Classification by Measurement

They range from simple category labels to precise numbers with a true zero point, dictating what math and tests you can use.

Scale	What the numbers carry	Operators	Examples
Nominal	labels only, no order	=, ≠	Color, City, Item Type, State
Ordinal	order, unequal gaps	=, ≠, >, <	Satisfaction (L/M/H), Rank
Interval	equal gaps, no true zero	+, -	Orders count, Time
Ratio	equal gaps and a true zero	×, ÷	Income, Temp, Duration

WHAT THIS DATA IS

Discrete vs Continuous

The set of possible values

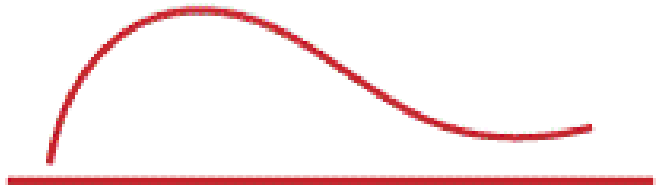
Discrete — countable, with gaps. Number of Students could be 71 or 72, not 70.2

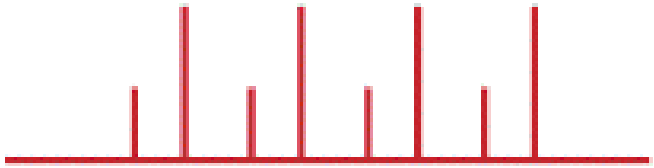
Continuous — any value in a range. The weight equal to 70.1 kg as precisely as you can measure.

What a probability can mean for discrete and continuous data

P(exactly 70 students) can be non-zero.

P(weight is exactly 70.1 kg) is **zero** — only intervals carry probability for a continuous variable.



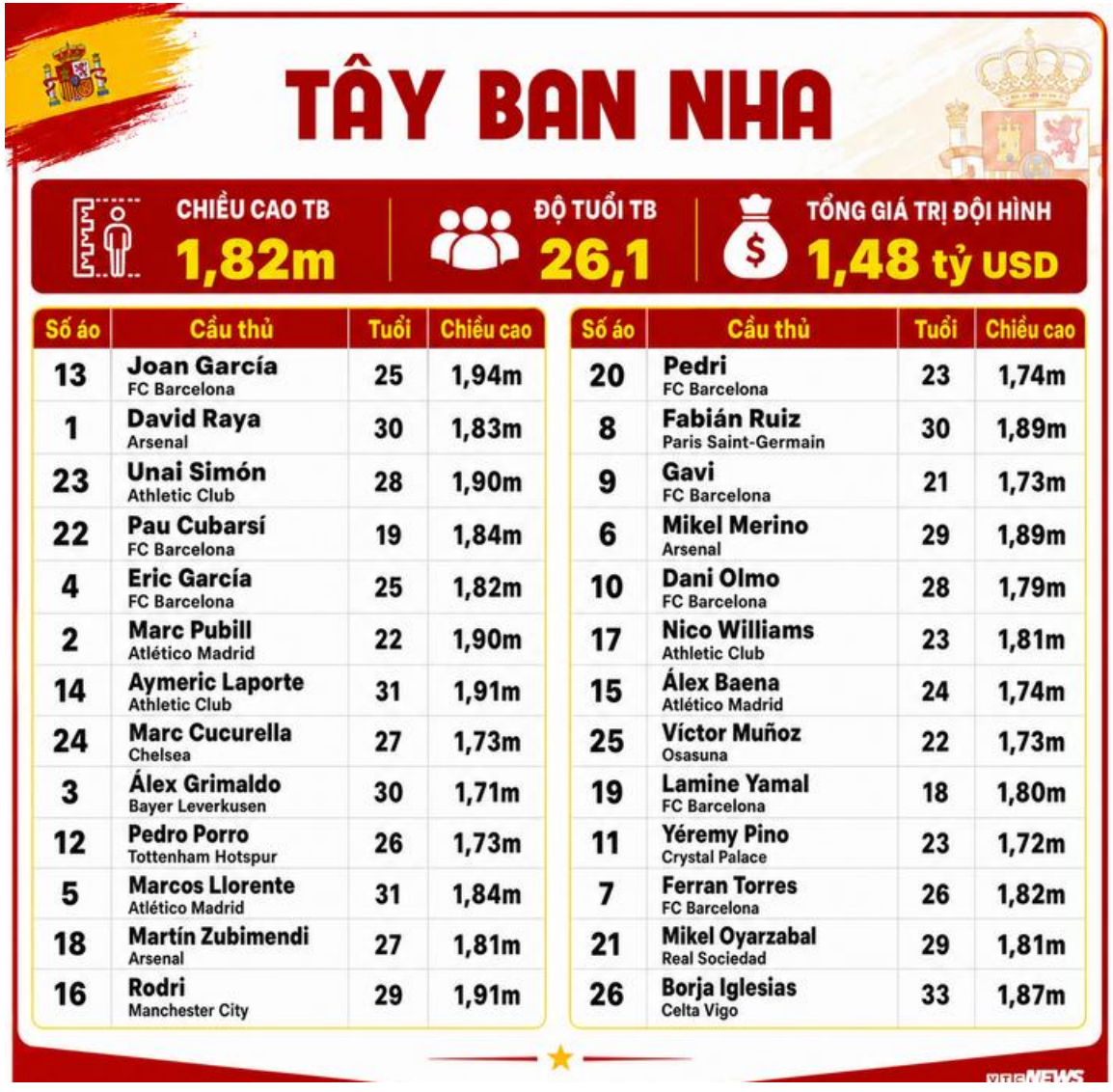


WHAT THIS DATA IS

How many statistics are there in the data?

How many numbers can you see?





WHAT THIS DATA IS

Throwback some Physics lesson

What are the difference between these 3 scales?



WHAT THIS DATA IS

Not all statistics are meaningful

Vote on each one before the verdict appears.

Mean of the Shirt number = 13.5

Meaningless. Nominal — the Shirt number are names wearing digits. Renumber them and the mean changes while the data does not.

Mean of a 1–5 GoogleMaps Review = 3.8

Contested. Ordinal — the gaps are not equal, so the sum has no fixed unit.

40°C is twice as hot as 20°C

False. Interval, no true zero. Switch to Kelvin and the ratio changes. Correct statement: 20 degrees warmer.

Change the labels, keep the data. If the statistic moves, it was never measuring the data.

Three ways for naming the centre

None of the three is the **AVERAGE**. They answer three different questions.

Statistic	How to calculate	Note
mean	Total sum divided by how many numbers there are	Can be misleading by outliers or skewed data
median	Ranking each value, and then take the exact middle	Ignores the actual values of data points and changes at the 2 ends
mode	Most frequent number appears	Can represent a “tiny minority”, ignore the rest of the data
all three	answer three different questions	the mode is the only centre that works on nominal data

Write these three words down: magnitude, rank, repetition. The rest of the session follows from them.

THE CENTRE

Bill Gates enters a bar

There are 9 men sitting in a bar, each earns $1000 a month. Suddenly, Bill Gates moves in, earning $100,000,000 a month. What is the average income in that room now — and are the 9 men any richer?

HOW IT WAS MADE

Sum the 10 incomes, divide by 10

Roughly 10 millions USD per person per month. The arithmetic is completely correct

WHAT IT CANNOT SAY

Nothing about how the 9 men live

One extreme value dragged the average away from every single person in the room.

In this example, calculate again using the mean, median and mode.

Which method is best used in this case?

SPREAD AND POSITION

Four steps to a standard deviation

Squaring is a choice, not a necessity — and it is exactly where fragility enters.

STEP 1

Deviation

Distance of each value from the mean.

STEP 2

Square each one

Otherwise they cancel to exactly zero, by definition of the mean.

STEP 3

Average the squares

That is the **variance**. Its unit is the original unit squared.

STEP 4

Take the square root

That is the **standard deviation**, back in the original unit.

$$\sigma = \sqrt{\frac{\sum(X-\mu)^2}{n}}$$

A deviation twice as large contributes four times as much.
SD is more sensitive to extremes than the mean is.

SPREAD AND POSITION

Which offer do you choose?

Both average 20 million VND a month. Offer A is fixed salary. Offer B is commission, SD about 12 million.

Option A: [20, 20, 20, 20, 20, 20, 20, 20, 20, 20, 20]
Mean: 20
SD: 0

Option B: [5, 8, 10, 38, 7, 35, 12, 15, 42, 8, 15, 45]
Mean: 20
SD: 12

There is no correct answer

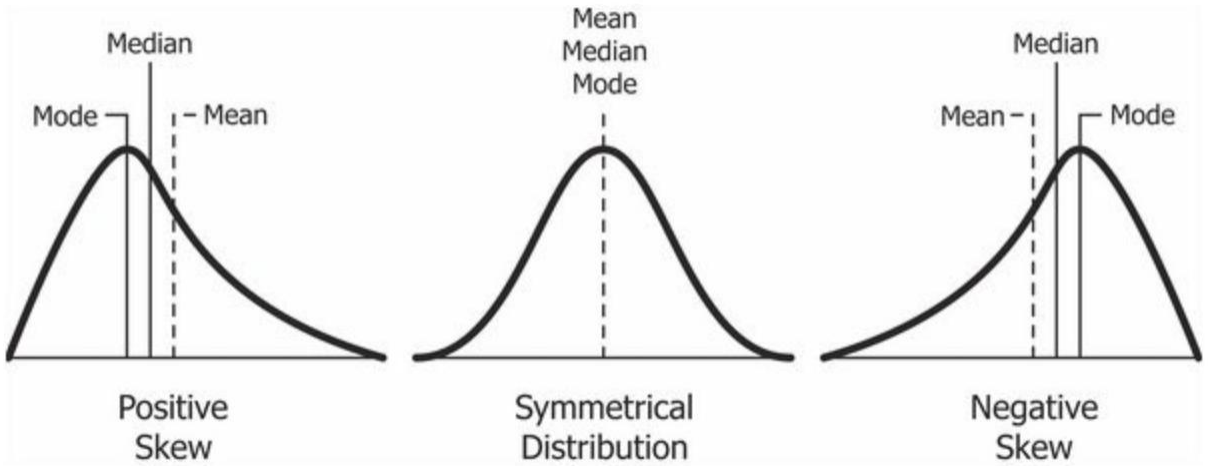
- The mean was **identical**, so the mean could not participate in the decision.
- The SD is the entire content of the choice.

In economics, spread is not noise around the answer. Spread is risk, and risk is the difference

Skewness

Asymmetry of a data distribution around its mean

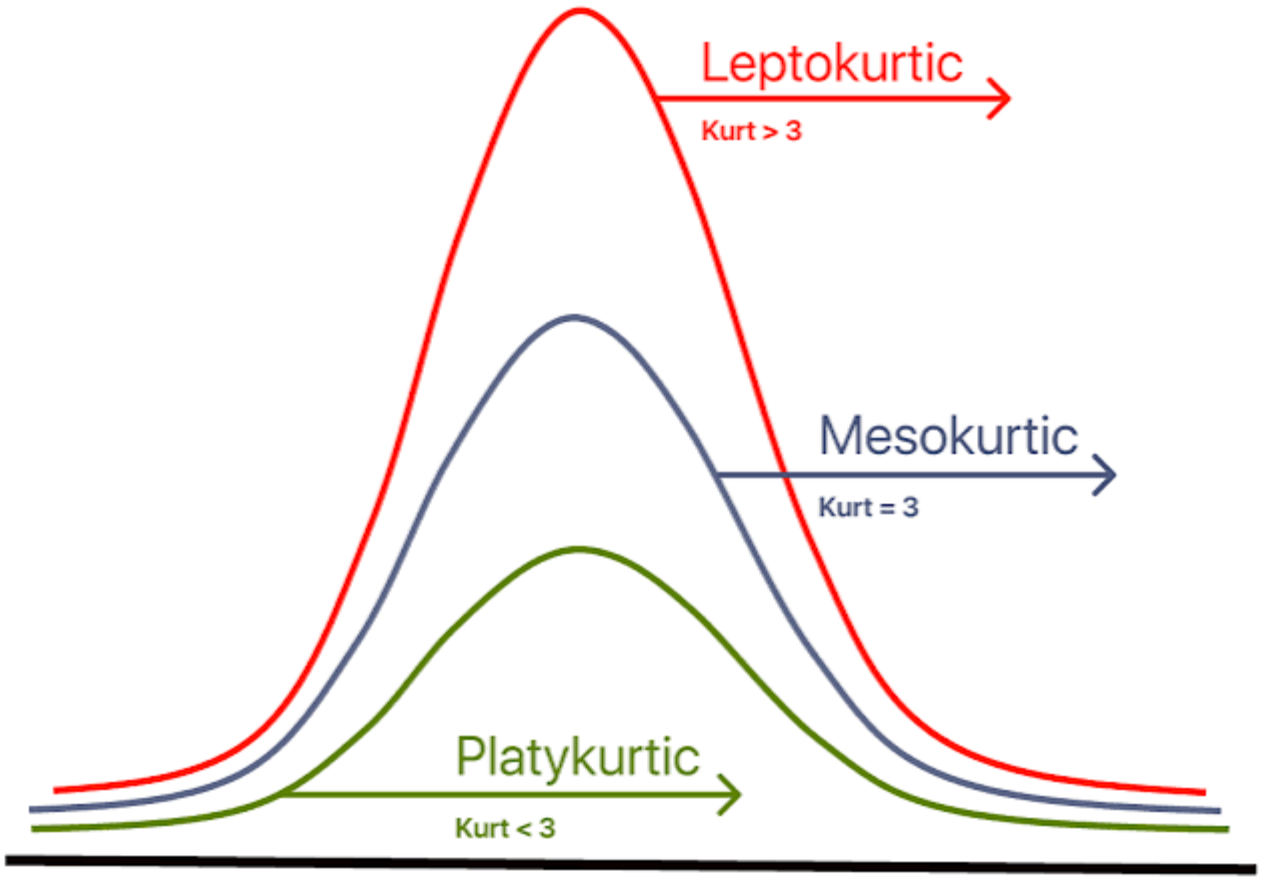
$$skewness = \frac{(x - \mu)^3}{\sigma^3}$$



Kurtosis

Measure that describes the "tailedness" of a probability distribution.

$$kurtosis = \frac{(x - \mu)^4}{\sigma^4}$$



SPREAD AND POSITION

Why economic variables lean right

A floor, no ceiling

Revenue, assets and headcount cannot go negative, so the left tail is walled. Nothing caps them above. And firms grow by **percentages**, not by additions.

- Repeated multiplication spreads the upper end further at every step.
- Most units bunch low, a thin tail stretches high. This is the **default** shape for size variables, not an anomaly.

Skew leaves a signature in the table

mean > median → right skew. **mean < median** → left skew. **mean ≈ median** → roughly symmetric.

- The reported skewness puts a sign on the same fact: **positive = right tail**. Read the sign; do not memorize thresholds for the magnitude.
- So one line of a summary table tells you the shape of a column you have not plotted. Session 3 draws it, Session 5 names it.

mode < median < mean is a signal of a right tail.

SPREAD AND POSITION

Log rescales the axis, and the shape follows

It turns ratios into equal distances

On the raw axis, 1 → 10 billion and 100 → 110 billion are the same gap — but one is a tenfold jump and one is 10%. On a log axis every tenfold jump is the same width.

- The right tail was built by repeated multiplication, so an axis that measures multiplication straightens it.

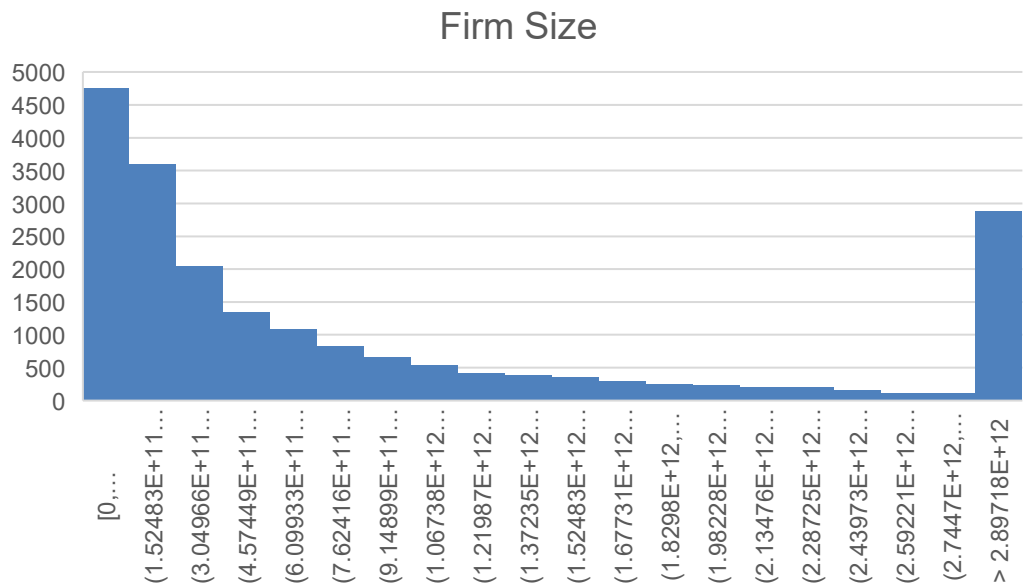
The mean of the logs is a different mean

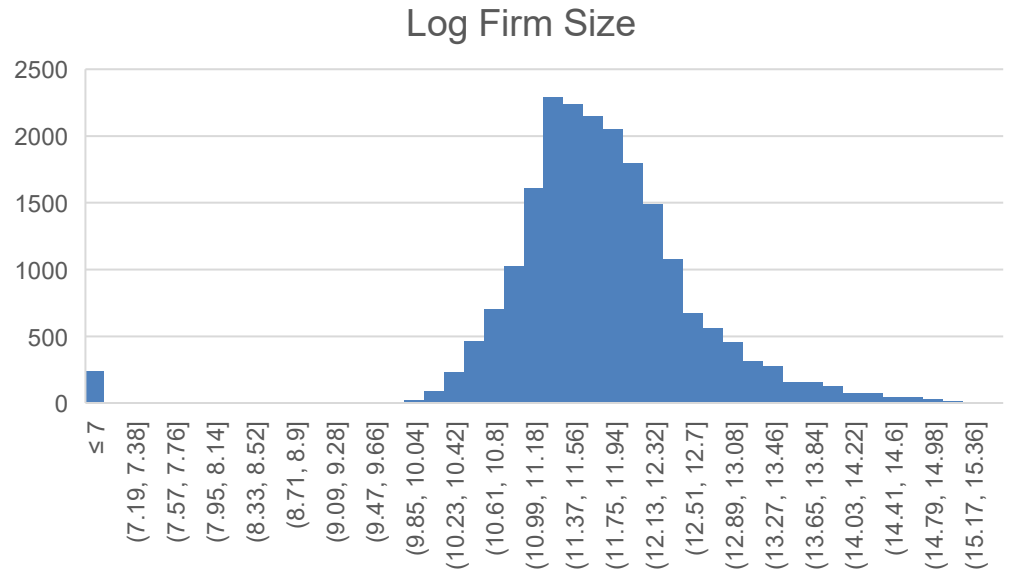
It is the **geometric mean** of the original values, not the arithmetic mean.

Log is not a repair. It is a change of measuring stick.

Logarithm rescales the axis, and the shape follows

Transforming the FirmSize to log(FirmSize)





Coefficient of Variation: spread without a unit

SD ÷ mean — the units cancel

It answers what SD cannot: relative to its own typical size, which variable moves more?

Firm Size has an SD in billions of VND, roa in percentage points.

- Same logic for exam scores on a 10-point scale versus a 1600-point scale.

The mean's problems become the CV's problems, and they arrive without any warning in the output.

- **Mean near zero** → the CV explodes.
- **Mean negative** → uninterpretable.

$$CV = \frac{\sigma}{\mu}$$

Rule for this course: CV only on ratio-scale variables with a comfortably positive mean.

SPREAD AND POSITION

Percentiles and IQR: spread built from rank

Sort the values. Q1 is the 25th percentile, Q3 the 75th. IQR = Q3 − Q1: the width of the middle half.

	Mean & SD	Median & IQR
Input used	every magnitude	ranks only
One extreme value	pulls both	moves neither
Feeds later sessions	almost all inference, Sessions 6–13	boxplots and outlier flagging, Session 3
Fails when	the tail is long	you need to do algebra with it

Neither pair is safer. They fail in different directions — which is exactly why a table reports both.

SPREAD AND POSITION

The z-score: turning a value into a position

You scored 7 in Mathematics and 8 in Literature. Which subject are you stronger in?

Mathematics — you scored 7

Class mean 5, SD 1. z = (7 − 5) ÷ 1 = **+2.0**. Two standard deviations above your class.

Literature — you scored 8

Class mean 8, SD 1. z = (8 − 8) ÷ 1 = **0.0**. Exactly average.

What z cannot do yet

z = +2.0 does **not** yet mean *top 2.3%*. That needs a distribution, and it arrives in Session 5.

The higher raw score is the weaker performance. The position was the information, not the number.

HANDS-ON

The full summary of one column

Every line answers a different question about the same 400 numbers.

Statistic	The question it answers	Value
N	how many units am I describing?	
mean	where is the balance point?	
median	what happens to the middle unit?	
SD	how far do units sit from the centre?	
min / max	what are the extremes?	
Q1 / Q3	where does the middle half begin and end?	
skewness	which tail is long?	

Read down this column and you can state the shape of a variable you have never plotted.

HANDS-ON

Excel and Python Functions

Type each Excel function by hand once. Every function you type is one concept from today.

Concept	Excel	Python
present values / missing	`=COUNT()` / `=COUNTBLANK()`	`.count()` / `.isna().sum()`
mean / median	`=AVERAGE()` / `=MEDIAN()`	`.mean()` / `.median()`
standard deviation	`=STDEV.S()`	`.std()`
Q1 / Q3	`=QUARTILE.INC(r,1)`	`.quantile([.25,.75])`
min / max	`=MIN()` / `=MAX()`	`.min()` / `.max()`
skewness / CV	`=SKEW()` / `=STDEV.S(r)/AVERAGE(r)`	`.skew()` / `.std()/.mean()`
all of it, by group	PivotTable	`.groupby('industry').describe()`

`COUNT` ignores blanks — which is exactly why comparing COUNT across columns exposes missingness.

Main Takeaways

1. COMPRESSION

Every statistic discards information by rule.

2. WAYS TO MEASURE CENTRE

Mean, Median, Mode

3. WAYS TO MEASURE SPREAD

Standard Deviation, Coefficient of Variation, Skewness, Kurtosis

4. LOG TRANSFORMATION

Why economics data often skew right, and why use log transformation

5. OVERALL DESCRIPTION OF DATA

Which metrics do we use to capture the basic of a dataset

6. PRACTICING WITH EXCEL

Excel Functions to perform descriptive analysis